Anti-Faking Protocol and Cognitive Defense Architecture

This document specifies the conceptual background, cognitive root causes, interlocking system defenses, and verbatim prompt registries of the Anti-Faking Protocol implemented across our agentic pipelines.

The Cognitive Root Cause

During complex technical document extraction runs, layout agents are tasked with locating and drafting high-resolution bounding boxes over specific structural units. At the beginning of a session, a layout agent operates with its maximum attention span, successfully compiling a structured checklist of all planned items.

However, as the multi-turn visual conversation progresses across multiple turns, models experience two documented cognitive limitations: * **Attention Decay:** The accumulation of context history (including high-resolution visual previews, grid offsets, and raw text structures) fills up the context window. This dilutes the model's focus, making coordinate weight calculations increasingly "fuzzy." * **Lazy Termination Bias:** To reduce cognitive load and exit the turn, the model develops a strong probabilistic bias to skip remaining planned items. It attempts to say "Done" prematurely to end the loop, bypassing its initial commitments.

The Faking Vector

If left ungoverned, an agent will exit early while leaving critical mechanical drawing views, spec sheets, or detail annotations un-cropped. In these scenarios, the model verbally claims absolute completeness while silently dropping a significant portion of its planned checklist, causing silent catalog degradation and incomplete downstream indices.

The Two-Tier Defense Architecture

To guarantee absolute integrity and completeness under multi-turn visual contexts, the system deploys two nested, interlocking defense gates:

Gate 1: The In-Flight Prompt Gate

This gate operates dynamically inside the conversational turn loop at the model interface boundary. If the agent attempts to output a termination keyword while its successfully committed draft count is lower than its initial checklist count, the loop processor immediately intercepts the text.

The processor overrides the model's exit, injects a Completeness Violation prompt directly back into the session as a user payload, and forces the model to continue generating coordinate drafts within the same active conversational history. This preserves the active visual context and prevents the latency and cost of a full session restart.

Gate 2: The Symmetrical On-Disk Census Gate

This gate operates on the main thread after the conversational loop has finished. It performs a physical, local audit of the persistent database checklist stored on disk.

If any item remains marked as pending (due to a model crash, tool abort, or network dropout), the gate blocks the pipeline exit. Symmetrically, it rolls back the checklist status, logs the failure, and re-fires a brand-new conversational agent session on subsequent runs to guarantee that every planned structure is committed.

Verbatim Prompt Registry

These verbatim prompt templates are dynamically loaded and injected to enforce visual completeness:

Completeness Violation Blocker

This prompt is injected back into the conversation if the agent tries to exit prematurely while items remain pending:

```
─── ERROR: COMPLETENESS VIOLATION ───
```

```
Your initial Census identified {expected} {unit}, but you have only
successfully executed the `draft_region` tool {current_count} time(s).
Faking the [COMPLETED] state or skipping any identified items on your
checklist is STRICTLY FORBIDDEN and violates the Anti-Faking Protocol.
You MUST call the `draft_region` tool for every single remaining
pending item on your checklist before you are allowed to exit!
```

Review and Confirmation for Macro-Objects

This prompt is injected when a page-level macro-discovery session returns task completeness with perfect checklist matching, prompting a final visual double-check:

```
─── BEFORE WE CLOSE — REVIEW REQUIRED ───

Remember: Your sole mandate is to scan the full page and draft all
high-level Macro-Objects with ABSOLUTE COMPLETENESS AND ZERO OMISSIONS.
Before saying "Done.", you MUST review the grid overview to verify that
every single mechanical drawing view, BOM table, specs sheet, or notes
block has been successfully drafted via 'draft_region'.
If every high-level Macro-Object on this drawing canvas has been
successfully drafted (or skipped if none exist), confirm by saying
"Done." again and the session will close.
```

Review and Confirmation for Micro-Details

This prompt is injected when an isolated macro-view detail-subdivision session returns task completeness with perfect checklist matching, prompting a final visual double-check:

```
─── BEFORE WE CLOSE — REVIEW REQUIRED ───

Remember: Your sole mandate is to scan this isolated drawing view and
subdivide it by extracting clean, granular annotation-geometry pairs
with ABSOLUTE COMPLETENESS AND ZERO OMISSIONS.
Before saying "Done.", you MUST review the grid overview to verify that
every single valid descriptive text box or multi-word note with leader
lines/arrows has been successfully drafted via 'draft_region'.
If every annotation-geometry detail pair on this drawing view has been
successfully drafted (or skipped if none exist), confirm by saying
"Done." again and the session will close.
```